



Startup Zone

- 1 A Fully-Electric Performance Motorcycle
- 2 How The IoT Is Being Utilised For A Cleaner India
- 3 AI-Based Intelligent Baggage Scanner System
- 4 Leap Motion Controller
- 5 Technology To Save Water And Boost Immunity

1 A FULLY-ELECTRIC PERFORMANCE MOTORCYCLE



(Credit: news.maxabout.com)

Why do many people hesitate to go electric? The main reason is range. What exactly is range? Just like mileage is distance travelled per litre of fuel, range of an electric vehicle (EV) is total distance driven on one full charge of the battery pack. And just like mileage of an internal combustion engine (ICE), range of an EV is also affected by circumstantial and environmental changes, such as city versus highway, road conditions, age of vehicle, operating temperatures, average weight riding and so on.

However, reasons for worry are different. In the case of ICEs, people worry about mileage because of the cost of fuel. In the case of EVs, they worry about range because of the fear of getting stranded, far away from a charge point.

There is a wide range of electric two-wheelers already out in the

market, and these vary according to type (scooter or motorcycle), weight, battery pack size and motor power. In terms of lifetime, two-wheeler EVs are already cheaper to own and run than petrol two-wheelers.

EVs are also more efficient than petrol or diesel vehicles in converting stored energy for driving. Add to this the lower cost of electricity versus petrol or diesel per kilometre. Regenerative braking in EVs can recover as much as one-fifth of the energy normally lost during braking. This results in fewer number of moving parts in EVs, leading to fewer breakdowns and servicing costs.

There are different factors that affect range of an EV. Aggressive driving reduces range, while smooth accelerations and decelerations help increase range. Non-stop high speeds reduce range as energy demand from the motor to the battery is high.

Lights also impact range as headlamps also consume energy. Running the vehicle under very hot temperatures can also affect range. It is important to ensure the vehicle/battery pack manufacturer has included adequate thermal protection.

Orxa Energies has designed Mantis, a fully-electric performance motorcycle, and the associated energy ecosystem to fit seamlessly into your lifestyle. It comes with decisive power and control for a stress-free ride. Speed and range allow you to be comfortable on all types of city roads and/or highways.

They have also built powerful battery packs that manage the specific challenges of performance vehicles, namely, longevity, thermals, weight and volume. This battery technology has already been deployed in multiple applications of aerospace and automotive. There are specific use-cases for both swap and charge, based on utilisation and per day kilometres.

Ranjita Ravi and Dr Prajwal Sabnis, co-founders, Orxa Energies, say, "Mantis will give riders 200 kilometres of range on real riding conditions with a massive 9kWh battery pack. Its top speed may reach up to 140kmph with 25kW motor power. It has a charging time of less than 3.5 hours, and a battery swap time of less than three minutes. Its targeted price is between ₹ 300,000 and ₹ 400,000, while the final price will be announced at the commercial launch in mid-2020s."

There are more and more people becoming aware of the value and benefits of driving an EV over an ICE vehicle, and governments the world over are pushing for measures to prolifer-